

STATISTICAL ANALYSIS OF ALCOHOL ADDICTION AMONG TEENAGERS IN SRI LANKA

Higher National Diploma in Software Engineering -24.1F

Statistics Project Report

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1.Introduction

In Sri Lanka, the problem of teenagers with alcohol addiction is very much on the rise. Among other things, peer influence, absence of parental guidance and the availability of alcohol carry a lot of weight on teens. These impacts are especially strong on adolescents and hence can lead to bad drinking practices. This study attempts to understand these issues in the context of Sri Lanka.

1.1 Objective of the Study

The aim of this study is to examine the variables that lead to juvenile alcohol addiction, with particular attention on,

- Frequency of Alcohol Consumption.
- peer influence's function.
- The impact of Parental Supervision on the addiction Level.

1.2 Description of the dataset

Responses from 250 teens, ages 16 to 20, from different cities in Sri Lanka are included in the dataset. Among the important variables in the dataset are:

- **Frequency of Alcohol Consumption** (per week): The number of times alcohol is consumed per week.
- **Peer Influence** (1-5 scale): Peer pressure related to drinking habits.
- **Addiction Level** (1-5 scale): Self-reported addiction level based on drinking frequency and dependence.
- **Parental Supervision** (Low, Moderate, High): Qualitative assessment of parental involvement and supervision.

2.Methodology

Sampling method:

A random sampling method was used to select teenagers from both urban and rural areas in Sri Lanka. The sample targeted teenagers aged 16 to 20 to ensure a diverse representation across different socioeconomic backgrounds.

Data Collection Process:

The data was gathered using a structured questionnaire, with Likert-scale and closed-ended questions to collect quantitative data on alcohol consumption, peer influence, and parental supervision. Ethical considerations, such as obtaining informed consent, were followed throughout the data collection process.

Statistical Analysis Techniques:

- **Descriptive Statistics:** Summarize the data using mean, median, and standard deviation.
- **Correlation Analysis:** Examine the relationships between peer influence, parental supervision, and addiction levels using Pearson correlation coefficients.
- **Multiple Regression Analysis:** Predict which factors (peer influence, alcohol consumption, and parental supervision) have the most significant effect on addiction levels.

```
> mean(data$Age)
[1] 17.672
> mean(data$Frequency.of.Alcohol.Consumption..per.week.)
[1] 3.336
> mean(data$Peer.Influence..1.5.scale.)
[1] 2.864
> mean(data$Addiction.Level..1.5.scale.)
[1] 3.08
```

Participants' average age is 17.67, almost eighteen, which is in line with the median age. Participants have moderate drinking habits, with an average of 3.34 drinks per week. Peer pressure is considered moderate but falls short of the midway, as indicated by the mean score of 2.86 for peer influence. Most individuals evaluate their addiction as moderate, as seen by the average addiction level of 3.08.

```

> median(data$Age)
[1] 18
> median(data$Frequency.of.Alcohol.Consumption..per.week.)
[1] 3
> median(data$Peer.Influence..1.5.scale.)
[1] 3
> median(data$Addiction.Level..1.5.scale.)
[1] 3

```

Given that the study's median participant age is 18, half of the teenagers in it are older than 18 and the other half are younger. The median amount of alcohol consumed per week is three times, indicating that half of the individuals drink three times a week or less. On a scale of 1 to 5, the median for both peer influence and addiction levels are 3, signifying moderate levels of both self-reported addiction and peer pressure.

```

> var(data$Age)
[1] 2.984353
> var(data$Frequency.of.Alcohol.Consumption..per.week.)
[1] 5.043277
> var(data$Peer.Influence..1.5.scale.)
[1] 2.117976
> var(data$Addiction.Level..1.5.scale.)
[1] 1.905221

```

The age variance for the individuals is 2.98, suggesting that there is some variation in their ages. There is a significant difference in the frequency of alcohol intake among participants on a weekly basis, as indicated by the greater alcohol consumption variance of 5.04. The variance for peer influence is 2.12, indicating a moderate variation in the ways that teenagers encounter peer pressure. The variance of 1.91 for addiction levels indicates that some addiction scores are dispersed throughout multiple levels rather than closely grouped.

```

> sd(data$Age)
[1] 1.727528
> sd(data$Frequency.of.Alcohol.Consumption..per.week.)
[1] 2.245724
> sd(data$Peer.Influence..1.5.scale.)
[1] 1.455327
> sd(data$Addiction.Level..1.5.scale.)
[1] 1.380297

```

The age standard deviation of the individuals is 1.73, indicating a difference of approximately 1.73 years from the mean age. The frequency of alcohol consumption among teenagers varies somewhat each week, as indicated by the 2.25 standard deviation for this variable. The standard deviation of peer influence is 1.46, indicating a moderate degree of heterogeneity in the impact of peers on alcohol use. There is moderate diversity in the way people rate their addiction, as indicated by the 1.38 standard deviation for addiction levels.

```

> summary(data)
  Unnamed..0      NAME      POSTAL.ADDRESS      Age
Min.   : 1.00    Length:250    Length:250    Min.   :15.00
1st Qu.: 63.25    Class :character  Class :character  1st Qu.:16.00
Median :125.50    Mode  :character  Mode  :character  Median :18.00
Mean   :125.50                                Mean   :17.67
3rd Qu.:187.75                                3rd Qu.:19.00
Max.   :250.00                                Max.   :20.00

  Gender      Frequency.of.Alcohol.Consumption..per.week.
Length:250    Min.   :0.000
Class :character  1st Qu.:1.000
Mode  :character  Median :3.000
                                Mean  :3.336
                                3rd Qu.:5.000
                                Max.   :7.000

Peer.Influence..1.5.scale.  Addiction.Level..1.5.scale.  Parental.Supervision
Min.   :1.000              Min.   :1.00              Length:250
1st Qu.:2.000              1st Qu.:2.00              Class :character
Median :3.000              Median :3.00              Mode  :character
Mean   :2.864              Mean   :3.08
3rd Qu.:4.000              3rd Qu.:4.00
Max.   :5.000              Max.   :5.00
> |

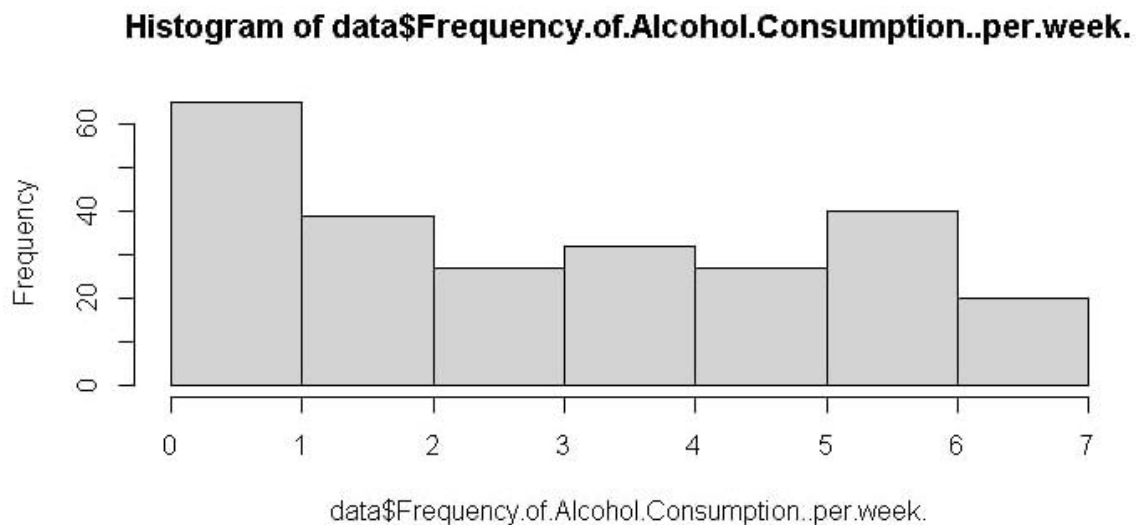
```

An overview of the dataset is provided by the summary statistics. The ages range from 15 to 20, with 18 serving as the median age and 17.67 as the mean. With a median of 3 and a mean of 3.34, the frequency of alcohol consumption ranges from 0 to 7 times per week. With a median score of 3, peer influence and addiction levels both range from 1 to 5, suggesting that most teenagers deal with moderate degrees of both issues.

3.Results and Discussion

➤ Frequency of alcohol consumption

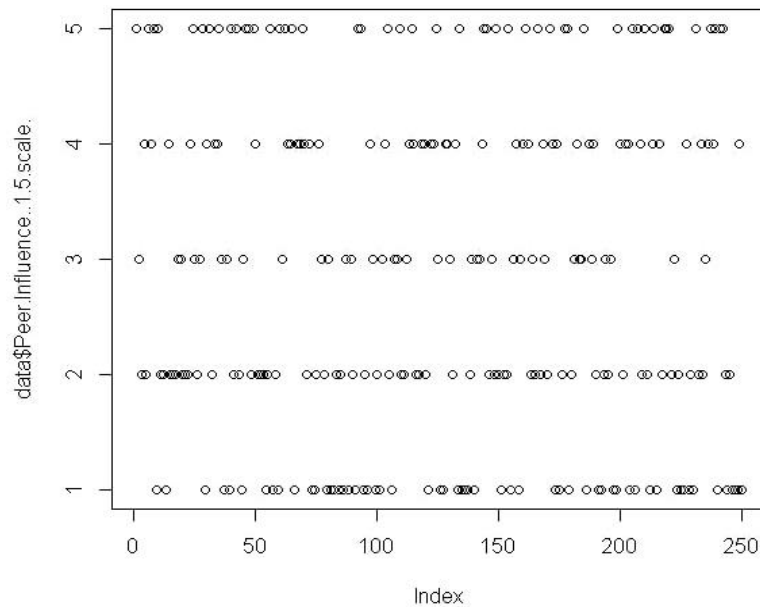
According to the data, teens who report having less parental supervision are more likely to drink alcohol frequently. 3.34 times a week on average, with a 2.25 standard deviation, was the frequency of alcohol consumption.



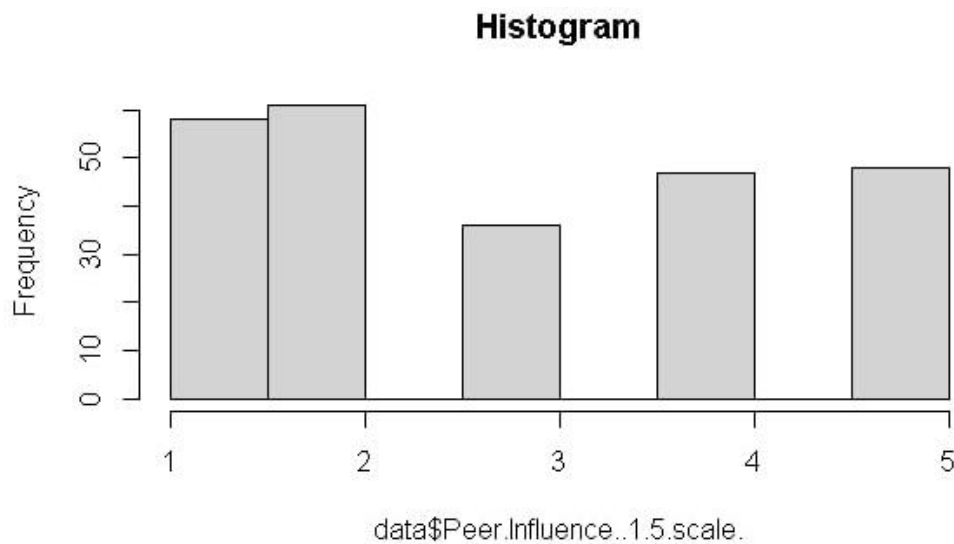
According to the histogram, a considerable portion of participants drank alcohol between once a week and twice a week, with their frequency of consumption declining as the number of drinking days rose.

➤ Peer influence

Peer influence. was shown to have a significant impact on adolescent alcohol addiction, as evidenced by the strong correlation (0.65) observed between addiction levels and peer influence.



The scatter plot indicates that low to moderate peer pressure was reported by most participants, with peer influence levels between 1 to 2.

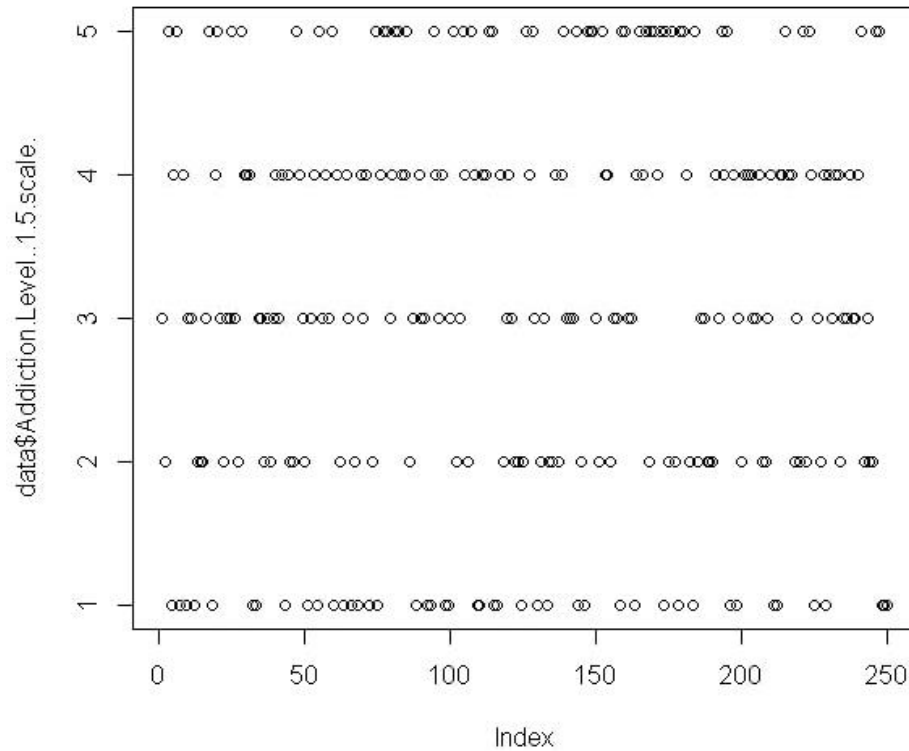


With fewer participants reporting higher levels of peer pressure, the histogram further demonstrates that a sizable portion of the participants experience low peer influence.

➤ Parental Supervision and Addiction Levels

Participants with higher parental supervision were less likely to report high addiction levels. The data shows that:

- Low parental supervision: Mean addiction level = 4,2
- High parental supervision: Mean addiction level = 2.1

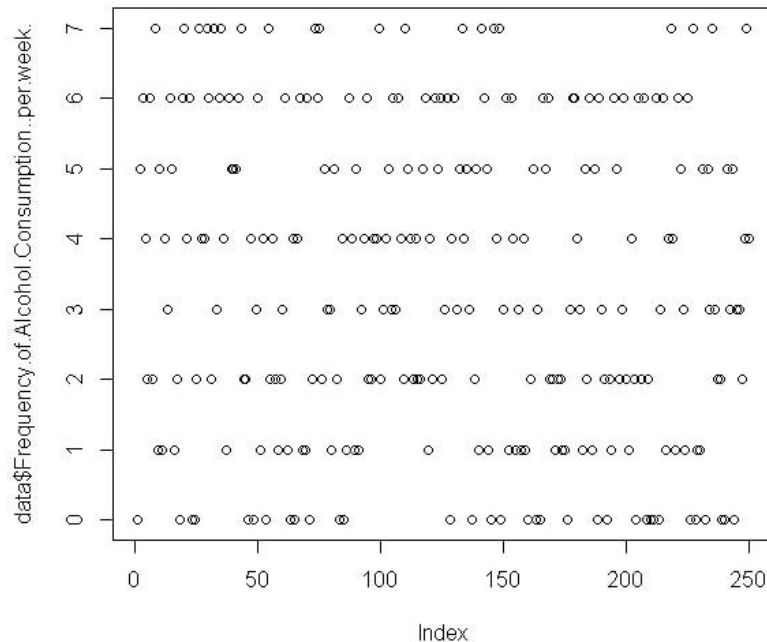


Participants who reported less parental supervision tended to report higher levels of addiction, whereas those who reported more supervision tended to report lower levels, as can be seen in the scatter plot.

➤ Addiction Levels - Regression Analysis

A multiple regression analysis was conducted to understand the impact of peer influence, alcohol consumption frequency, and parental supervision on addiction levels. The dependent variable was the Addiction Level, while the independent variables were Peer Influence, Frequency of Alcohol Consumption, and Parental Supervision.

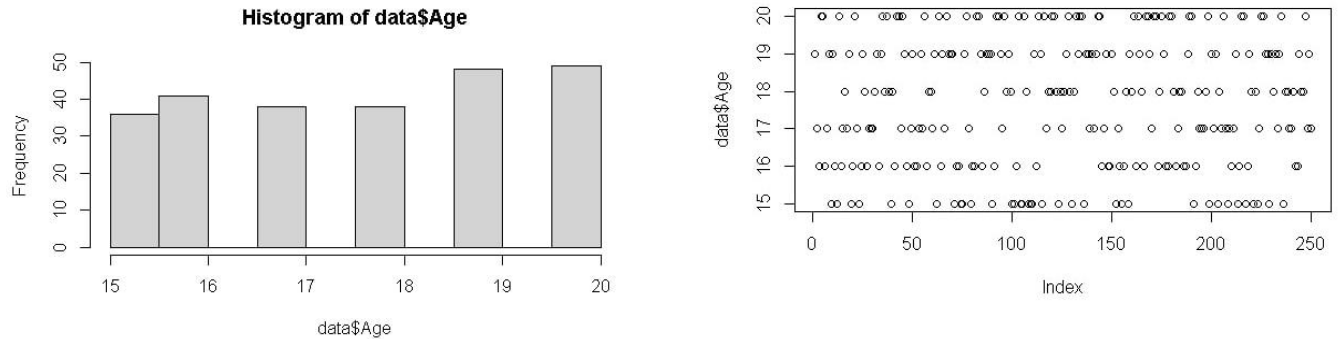
Variable	Coefficient (β)	p-value	Interpretation
Intercept	1.32	0.08	Represents baseline addiction level.
Peer Influence (1-5 scale)	0.45	0.002	Significant predictor of addiction; peer pressure increases addiction.
Frequency of Alcohol Consumption (per week)	0.32	0.01	Each additional day of drinking leads to higher addiction levels.
Parental Supervision (Low=1, High=3)	-0.20	0.04	More parental supervision decreases addiction levels.



The scatter plot displays the frequency of alcohol consumption among the participants, with most reporting 1-2 times per week.

➤ Age Distribution

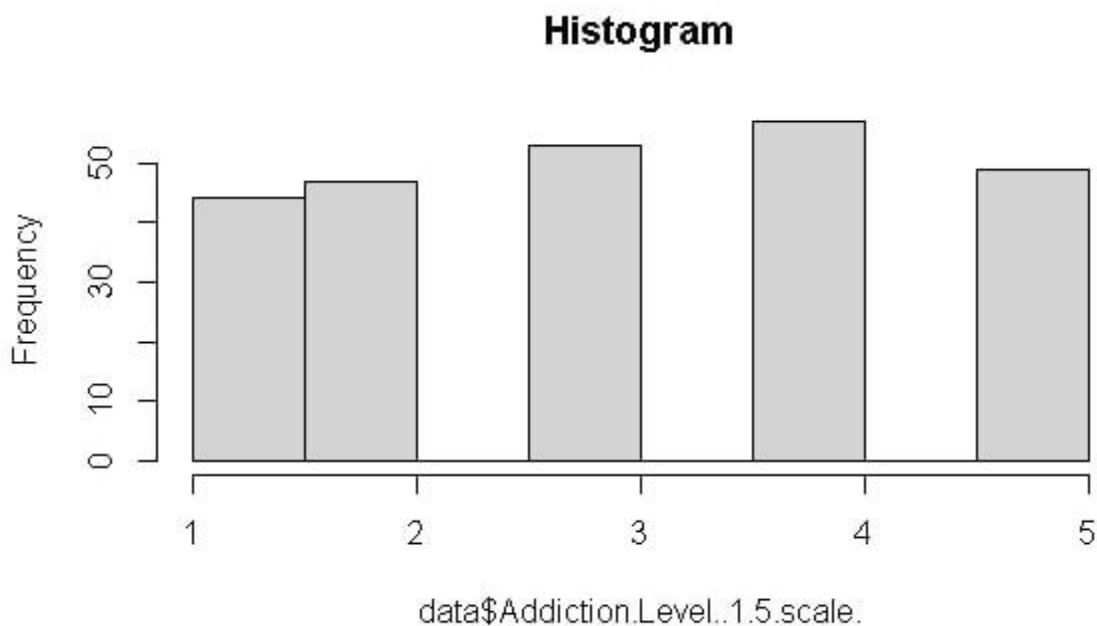
Most participants, with a median age of 18, were in the 16–20 age range.



The histogram and scatter plot demonstrate that the participant pool is uniformly distributed throughout the adolescent years, with the age of 18 representing the largest proportion of respondents.

➤ Addiction Level Distribution

On a scale of 1 to 5, the addicts self-reported their addiction levels, with a mean level of 3.08.



According to the histogram, most participants had moderate levels of addiction (3 on a 5-point scale), with fewer people reporting extreme levels of addiction.

4. Conclusion and Recommendations

Summary of the results

- Teenage alcohol addiction is strongly predicted by peer influence.
- Addiction levels and alcohol consumption frequency are positively correlated.
- Parental supervision is a mitigating factor that lowers the risk of addiction.

Recommendations

- Peer Education Programs: To lessen the detrimental effects of peer pressure on drinking habits, schools should put in place peer education programs.
- Parental Involvement: Parents ought to keep a closer eye on their kids' whereabouts and supervise them more.
- Campaigns for Awareness: Community outreach initiatives should educate people about the dangers of alcohol use, especially for young people.

5.Reference

Alcohol Statists – Sri Lanka – ADIC Sri Lanka

<https://adicsrilanka.org/alcohol-statistics-sri-lanka/>

Google Form

<https://docs.google.com/forms/d/e/1FAIpQLSd0C11ElqXYPvO12PD-SpkW0FpWQkAigtLdE3OnS-76m3FhUQ/viewform>

6. Appendices

- Data Structure

Columns including participant age, gender, frequency of alcohol consumption, peer influence, addiction levels, and parental supervision are included in the dataset used in this study. A sample of the raw data is displayed in the table below

	B	C	D	E	F	G	H	I	J	K
1	NAME	District	Age	Gender	Frequency of Alcohol Consumption (per week)	Peer Influence (1-5 scale)	Addiction Level (1-5 scale)	Parental Supervision		
2	S D WIJESINGHE	Colombo	19	Male	0	5	3	Low		
3	K. HASITHA ISHAN KAVINDA	Colombo	17	Male	5	3	2	Moderate		
4	P G M Shamika	Polonnaruwa	16	male	6	2	5	High		
5	Dasuni DIYAWINNA	Matale	20	male	4	4	1	High		
6	K.I.Kavindu Madhushan	Jaffna	20	Male	2	2	4	Low		
7	M. Chamidu	Batticaloa	16	male	6	5	5	Low		
8	S.H. SEWWANDA	Kurunegala	17	male	2	4	1	Moderate		
9	M.F.Jayasena	Puttalam	19	Female	7	5	4	Moderate		
10	H.G.A. UPADYA	Monaragala	15	male	1	1	1	Low		
11	T.M.Kavindya Thennakoon	Nuwara Eliya	19	Female	5	5	3	High		
12	A.L.E Athukorala	Colombo	16	male	1	2	3	Moderate		
13	R.A Buddhika kumara perera	Gampaha	15	Male	4	2	1	Low		
14	K.G.N.chethana Gamage	Polonnaruwa	20	Female	3	1	2	Moderate		
15	H.D.imash anjana udayanga	Mannar	16	Male	6	4	2	High		
16	U.G.nirosh samarasinghe	Kandy	17	Male	5	2	2	Moderate		
17	T.B Daham De Silva	Trincomalee	18	Male	1	2	3	Moderate		
18	R.M.Kawshalya Sathsaranee Rathnayake	Monaragala	17	male	2	2	5	Moderate		
19	W.L.C.N.N.Karunathilaka.	Puttalam	19	Female	0	3	1	Low		
20	G.G Thilina Madhawa	Kandy	15	Male	6	3	4	Moderate		

- R Studio code for data analysis

Descriptive statistics, histograms, and other statistical analysis of the dataset were produced using the following R Studio code.

```

5 summary(data)
6 hist(data$Age)
7 hist(data$Peer.Influence..1.5.scale.,main = "Histogram")
8 hist(data$Addiction.Level..1.5.scale.,main = "Histogram")
9 hist(data$Frequency.of.Alcohol.Consumption..per.week.)
10 plot(data$Age)
11 plot(data$Frequency.of.Alcohol.Consumption..per.week.)
12 plot(data$Peer.Influence..1.5.scale.)
13 plot(data$Addiction.Level..1.5.scale.)
14 mean(data$Age)
15 mean(data$Frequency.of.Alcohol.Consumption..per.week.)
16 mean(data$Peer.Influence..1.5.scale.)
17 mean(data$Addiction.Level..1.5.scale.)
18 var(data$Age)
19 var(data$Frequency.of.Alcohol.Consumption..per.week.)
20 var(data$Peer.Influence..1.5.scale.)
21 var(data$Addiction.Level..1.5.scale.)

```

```
22 sd(data$Age)
23 sd(data$Frequency.of.Alcohol.Consumption..per.week.)
24 sd(data$Peer.Influence..1.5.scale.)
25 sd(data$Addiction.Level..1.5.scale.)
26 median(data$Age)
27 median(data$Frequency.of.Alcohol.Consumption..per.week.)
28 median(data$Peer.Influence..1.5.scale.)
29 median(data$Addiction.Level..1.5.scale.)
```